Subject 4: Prediction of the Critical Temperature of Superconducting Materials

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Subject: Subject 4, prediction of the critical temperature of superconducting materials.

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1 Introduction and Objective

Superconducting materials are characterized by a critical temperature above which their superconducting behavior disappears. Predicting this temperature from chemical descriptors is a useful statistical learning problem because experimental measurement is costly, while composition-level information is comparatively easy to record. In this project, the response variable is $Y =$ `critical_temp`, measured in Kelvin, and the predictors are the 81 numerical physical and chemical covariates supplied in the UCI superconductivity dataset.

The problem is formulated as supervised regression. Given observations (x_i, y_i), with $x_i \in \mathbb{R}^{81}$ and $y_i \in \mathbb{R}_+$, the objective is to build a function $\widehat{f}$ such that $\widehat{f}(x)$ predicts the critical temperature of a new material as accurately as possible. We compare a constant baseline, ordinary least squares, Ridge regression, Lasso regression, k-nearest-neighbor regression, and a histogram-based gradient boosting regressor. The first four non-baseline methods allow us to connect the analysis with the course material, while the boosting model is included as an explicitly explained nonlinear benchmark.

2 Dataset and Data Audit

The raw UCI dataset contains 21,263 materials. Each material is described by 81 engineered numerical variables derived from chemical formulae and elemental properties such as atomic mass, first ionization energy, atomic radius, density, electron affinity, fusion heat, thermal conductivity, and valence. The engineered variables include summary statistics such as arithmetic means, weighted means, geometric means, ranges, entropies, and standard deviations. The target column is `critical_temp`.

Before modeling, exact duplicate rows in the full feature-target table are removed. This is done before the train-test split, so identical observations cannot appear once in the training set and once in the test set. The cleaning step removes 66 exact duplicates and leaves 21,197 observations. No missing values are present after cleaning, so no imputation policy is needed for this version of the dataset. The first-stage data audit is saved by the pipeline as JSON and CSV artifacts under `reports/tables/`.

The target distribution is right-skewed. The mean critical temperature is approximately 34.49 K, the median is 20 K, and the maximum is 185 K. This means that a model can obtain low average error on the many low-temperature materials while still making scientifically important mistakes in the high-temperature tail. For that reason, the model comparison is supplemented with residual diagnostics by target range.

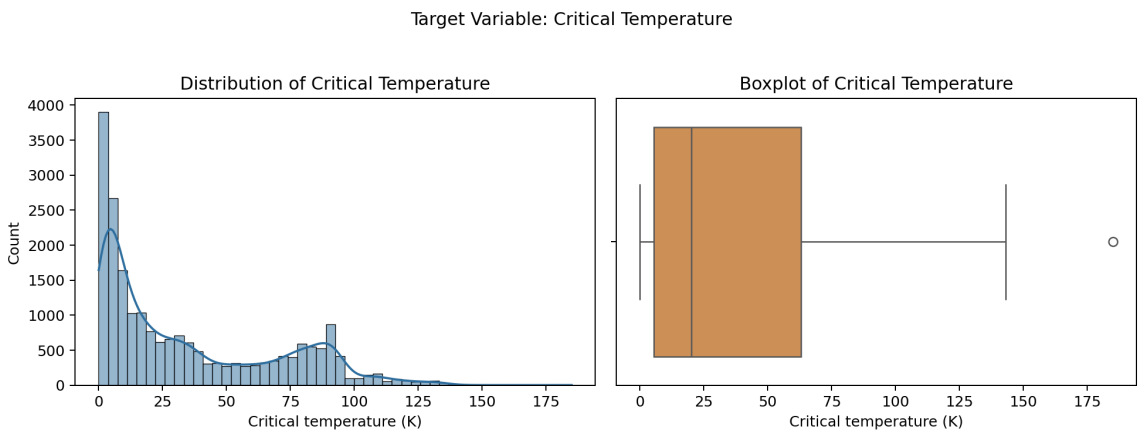

Figure 1: Distribution and boxplot of the critical temperature. The response is positive and right-skewed, with a long upper tail of high-temperature materials.

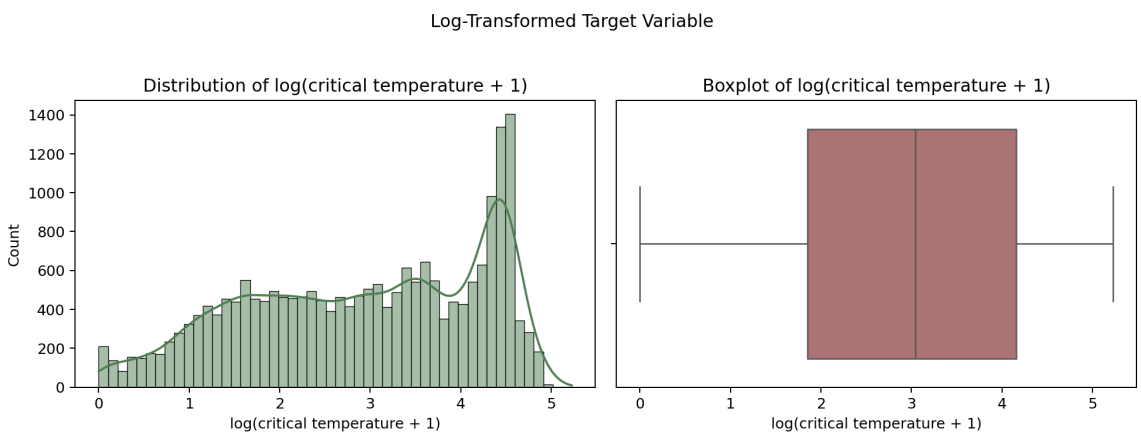

Figure 2: Distribution and boxplot after the transformation $\log(1 + \texttt{critical_temp})$. The transformed response is less skewed, which motivates using this transformation inside the modeling pipelines.

3 Exploratory Data Analysis

The exploratory analysis has three goals: to understand the target variable, to check the quality and structure of the covariates, and to identify possible nonlinear relationships that could influence model choice. The marginal Pearson correlations show that several thermal conductivity, atomic radius, valence, atomic mass, and ionization-energy descriptors are strongly related to the target. These correlations are not used as a feature selection rule; they are descriptive statistics that help interpret the dataset and the behavior of later models.

The feature families have different levels of marginal association with the response. The strongest single correlation is obtained by `wtd_std_ThermalConductivity`; thermal conductivity, atomic radius, valence, atomic mass, and the number of elements all contain variables with

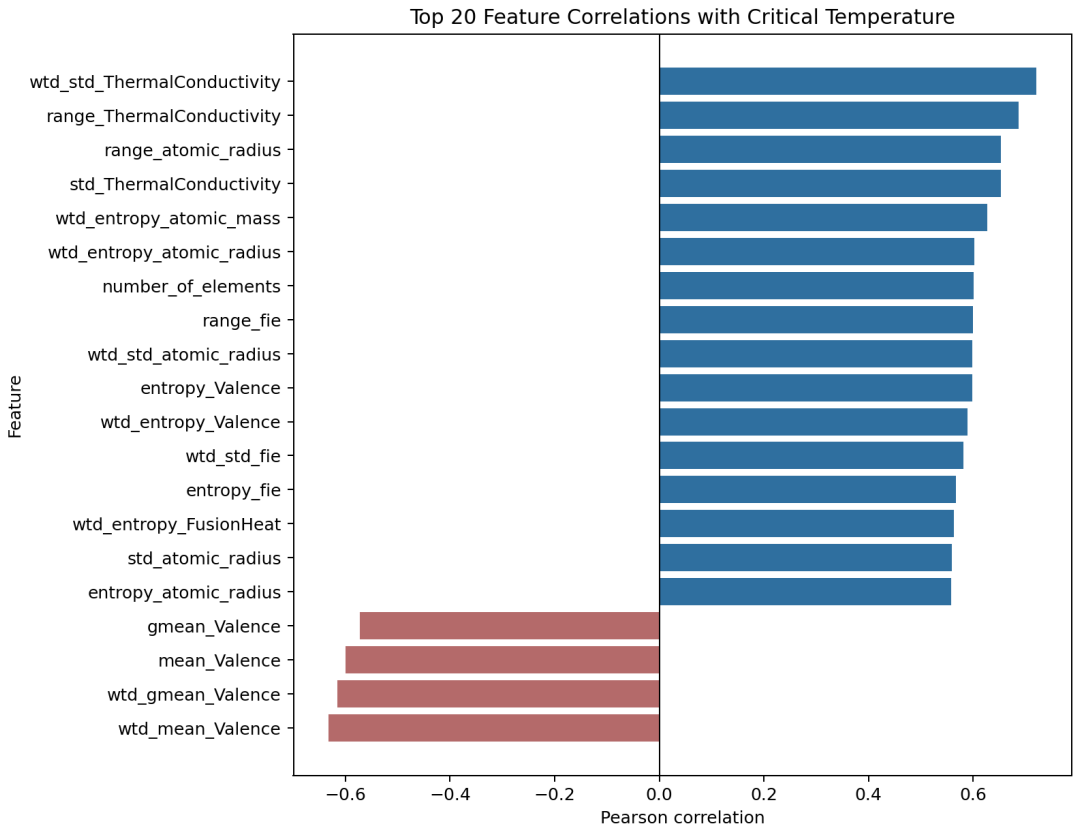

Figure 3: The twenty features with the largest absolute Pearson correlation with the critical temperature. The signs indicate whether the marginal linear relation is positive or negative.

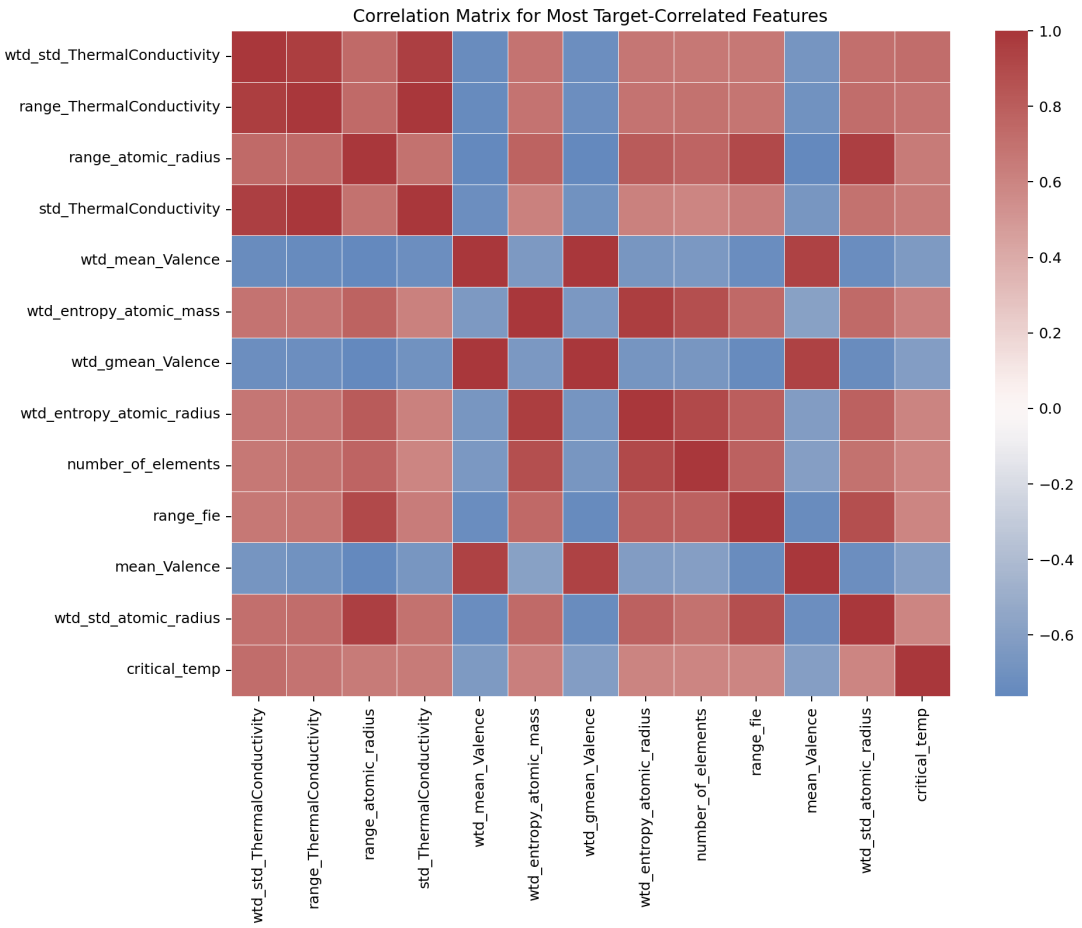

Figure 4: Correlation matrix for the features most correlated with the target. Strong dependence between covariates is visible, which supports the use of regularized linear models such as Ridge and Lasso.

absolute correlations above 0.60. These observations are consistent with the physical intuition that both composition complexity and elemental property variation matter.

Family	Features	Mean |corr|	Max |corr|	Strongest feature	Corr.
ThermalConductivity	10	0.425	0.721	wtd_std_ThermalConductivity	0.721
atomic_radius	10	0.427	0.653	range_atomic_radius	0.653
Valence	10	0.470	0.632	wtd_mean_Valence	-0.632
atomic_mass	10	0.376	0.627	wtd_entropy_atomic_mass	0.627
number_of_elements	1	0.600	0.600	number_of_elements	0.600
fie	10	0.384	0.600	range_fie	0.600
FusionHeat	10	0.361	0.563	wtd_entropy_FusionHeat	0.563
Density	10	0.361	0.541	gmean_Density	-0.541
ElectronAffinity	10	0.251	0.436	entropy_ElectronAffinity	0.436

Table 1: Correlation summary by engineered feature family. For each family, the table reports the average absolute marginal correlation and the strongest individual feature.

The number of chemical elements in the formula is itself an informative covariate. Single-, two-, and three-element materials are mostly low-temperature in this dataset, whereas materials with four or more elements are much more likely to reach high critical temperatures. The 77 K threshold is useful as a descriptive reference because it corresponds approximately to the boiling point of liquid nitrogen and is often used to distinguish practically attractive high-temperature superconductors.

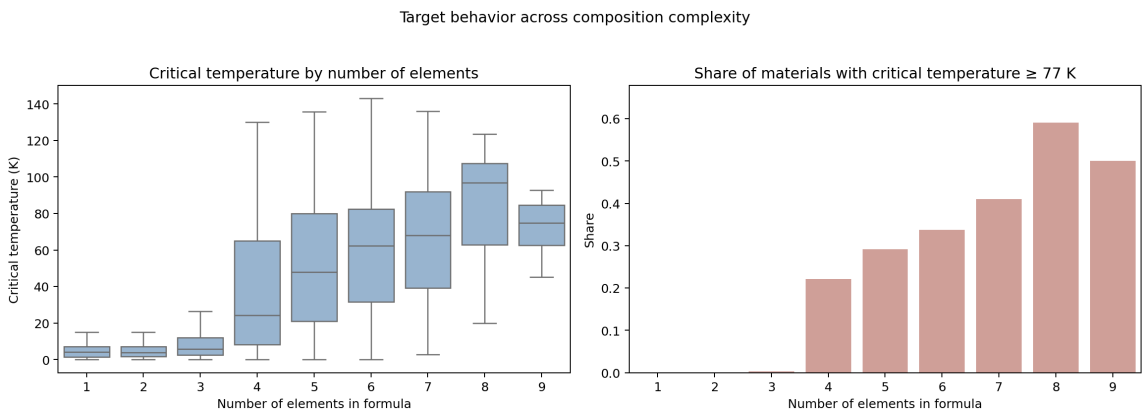

Figure 5: Critical temperature as a function of the number of elements. The boxplots show the shift in the target distribution, and the bar plot shows the proportion of materials with critical temperature at least 77 K.

4 Preprocessing and Leakage Control

The preprocessing sequence is designed to make the final test-set results credible. First, the raw UCI feature and target tables are merged and exact full rows are removed. Second, the cleaned table is split once into a training set and a held-out test set using an 80/20 split with random seed 42. The held-out test set is not used for hyperparameter selection.

For OLS, Ridge, Lasso, and k-nearest neighbors, the covariates are standardized inside a scikit-learn pipeline. The centering and scaling parameters are fit only on the relevant training fold during cross-validation and only on the full training set before final test evaluation. This avoids data leakage from the test set into preprocessing. The tree-based histogram gradient boosting model is not sensitive to monotone feature scaling, so it is fit without standardization.

Elements	N	Mean K	Median K	Q1 K	Q3 K	Min K	Max K	Share >= 77 K
1	285	5.188	4.000	1.300	7.100	0.000	52.000	0.000
2	3243	5.729	3.890	1.727	7.100	0.001	185.000	0.000
3	3869	9.347	5.740	2.500	12.100	0.010	132.000	0.003
4	4493	36.550	24.300	8.200	65.000	0.001	130.000	0.221
5	5792	50.714	47.900	20.988	80.000	0.001	135.800	0.291
6	2666	59.629	62.400	31.500	82.400	0.100	143.000	0.337
7	774	66.833	68.000	39.200	92.000	2.700	136.000	0.410
8	61	86.521	96.800	63.000	107.500	20.000	123.500	0.590
9	14	73.229	74.825	62.475	84.588	45.300	92.900	0.500

Table 2: Target summary by number of elements in the material formula. The share column reports the fraction of observations with critical temperature at least 77 K.

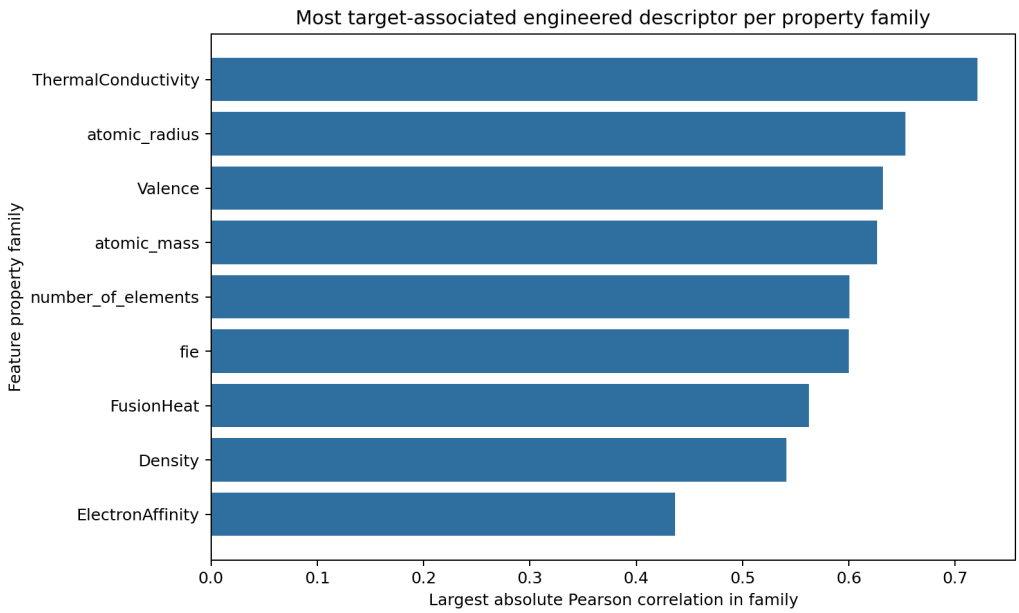

Figure 6: Largest absolute marginal target correlation within each engineered physical-property family. This complements the individual feature ranking by grouping related descriptors.

The target is transformed during fitting using

$$Z = \log(1 + Y).$$

Predictions are transformed back to the original Kelvin scale using $\widehat{Y} = \exp(\widehat{Z}) - 1$ before metrics are computed. The transformation reduces the dominance of the high-temperature tail during fitting and ensures that model comparisons are still reported in physically interpretable Kelvin units.

5 Models

Constant baseline. The baseline predicts a constant value learned from the training data. It provides a minimal reference: a useful model must improve substantially on this baseline.

Ordinary least squares. The linear regression model assumes $\widehat{f}(x) = \widehat{\beta}_0 + x^\top \widehat{\beta}$, where the coefficients are chosen to minimize the residual sum of squares on the transformed target. OLS is simple and interpretable, but it can be unstable when predictors are correlated.

Ridge regression. Ridge regression adds an ℓ_2 penalty to stabilize the linear model:

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} (z_i - \beta_0 - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.$$

The penalty shrinks coefficients toward zero without setting them exactly to zero. The regularization parameter is selected by cross-validation on the training set.

Lasso regression. Lasso regression replaces the Ridge penalty by an ℓ_1 penalty:

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} (z_i - \beta_0 - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

This can set some coefficients exactly to zero, which makes the fitted linear model more sparse. As with Ridge, the penalty parameter is selected without using the held-out test set.

k-nearest-neighbor regression. k-NN predicts a new response by averaging the responses of the closest training observations in standardized feature space. It is nonlinear and can adapt to local structure in the data. The number of neighbors k is selected by grid search with cross-validation. This method is powerful here but stores the training set and can be sensitive to the choice of scaling and distance metric.

Histogram-based gradient boosting. The histogram gradient boosting regressor is included as an additional nonlinear benchmark. It builds an additive model of regression trees, where each new tree is fit to reduce the current prediction error. This method is not treated as a course-covered method in the report, so its functioning is stated explicitly. It is useful for this dataset because interactions between chemical descriptors are plausible and may not be captured by linear models.

6 Model Selection and Evaluation Criteria

Model selection is performed only with the training data. Ridge and Lasso choose regularization strengths from predefined grids, and k-NN chooses $k \in \{3, 5, 7, 11, 15, 21, 31\}$ by five-fold cross-validation. The gradient boosting hyperparameters are fixed in the configuration file and include a learning rate of 0.06, 350 maximum boosting iterations, 31 maximum leaf nodes, ℓ_2 regularization 0.01, and early stopping on an internal validation fraction.

For observations $(y_i, \widehat{y}_i)_{i=1}^n$ in the held-out test set, the main criteria are

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}},$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|, \qquad R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

RMSE is the primary selection criterion because it is expressed in Kelvin and penalizes large errors strongly. MAE is less sensitive to isolated large errors, and R^2 measures the fraction of test-set variance explained relative to a constant predictor.

7 Results

Model	MSE	RMSE	MAE	R2
knn_gridsearch	113.5725	10.6570	5.6747	0.9035
hist_gradient_boosting	115.4570	10.7451	6.2916	0.9019
ridge_cv	334.8801	18.2997	12.4752	0.7155
linear_regression	334.9093	18.3005	12.4742	0.7155
lasso_cv	336.2314	18.3366	12.5203	0.7144
dummy_regressor	1469.0342	38.3280	27.6689	-0.2480

Table 3: Held-out test performance. Hyperparameters are selected on the training data, and this table is computed only on the test set.

The nonlinear models outperform the linear family by a large margin. The k-nearest-neighbor model obtains the lowest RMSE, approximately 10.66 K, the lowest MAE, approximately 5.67 K, and the highest R^2, approximately 0.904. The histogram-based gradient boosting model is very close, with RMSE approximately 10.75 K and R^2 approximately 0.902. OLS, Ridge, and Lasso remain around RMSE 18.3 K. This gap suggests that the relationship between the engineered physical descriptors and the critical temperature contains nonlinear structure or interactions that linear models do not capture well.

The best selected k-NN model uses $k = 3$, while Ridge selects a small penalty $\alpha = 0.1$ and Lasso selects $\alpha = 10^{-4}$. The small amount of regularization selected for the linear models is consistent with the fact that the linear signal is present but insufficient: shrinkage stabilizes coefficients, yet it cannot create the nonlinear local behavior captured by k-NN and boosting.

The residual diagnostics indicate that the selected model has little average bias overall, with a test residual mean of about 0.39 K. However, the error is not uniform over the target range. Predictions are most accurate for low-temperature materials below 10 K. Errors are larger for the high-temperature tail, especially for materials above 100 K, where the model tends to underpredict on average. This is an important limitation because the high-temperature region is often the most scientifically interesting part of the dataset.

Model	N	Mean resid.	Resid. SD	Median resid.	Resid. q05	Resid. q95	MAE	RMSE
knn_gridsearch	4240	0.3865	10.6513	0.0687	-15.3671	16.2945	5.6747	10.6570
hist_gradient_boosting	4240	1.6589	10.6175	0.6028	-14.1505	17.9158	6.2916	10.7451
ridge_cv	4240	3.9722	17.8655	1.5179	-24.6372	35.7907	12.4752	18.2997
linear_regression	4240	3.9719	17.8664	1.5211	-24.6689	35.8508	12.4742	18.3005
lasso_cv	4240	4.0214	17.8923	1.5333	-24.6219	35.8532	12.5203	18.3366
dummy_regressor	4240	17.0847	34.3136	2.5313	-16.3687	77.5313	27.6689	38.3280

Table 4: Residual diagnostics by model on the held-out test set. Residuals are defined as observed minus predicted critical temperature in Kelvin.

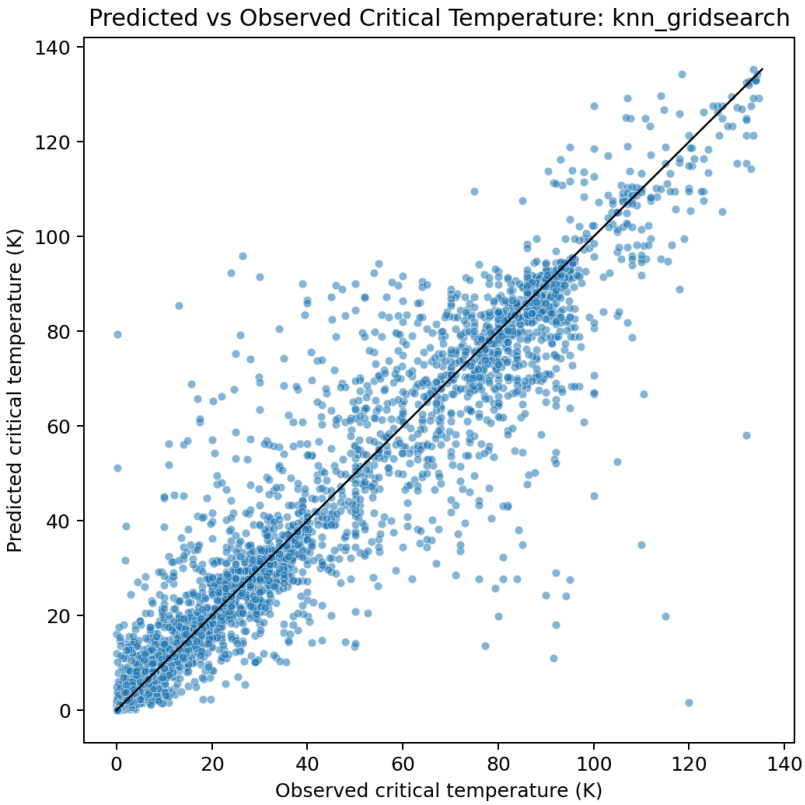

Figure 7: Predicted versus observed critical temperature for the selected model. Points near the diagonal correspond to accurate predictions on the Kelvin scale.

Model	Target range	N	Mean true K	Mean pred. K	Mean resid.	MAE	RMSE
knn_gridsearch	<10 K	1518	3.931	4.832	-0.901	1.767	4.186
knn_gridsearch	10-30 K	953	18.488	20.350	-1.862	5.410	10.060
knn_gridsearch	30-77 K	971	50.566	50.501	0.065	9.574	13.438
knn_gridsearch	77-100 K	636	87.008	81.443	5.566	8.095	13.838
knn_gridsearch	>=100 K	162	114.091	106.824	7.268	10.978	19.463

Table 5: Selected-model error by observed critical-temperature range. The table highlights that test error increases in the high-temperature tail.

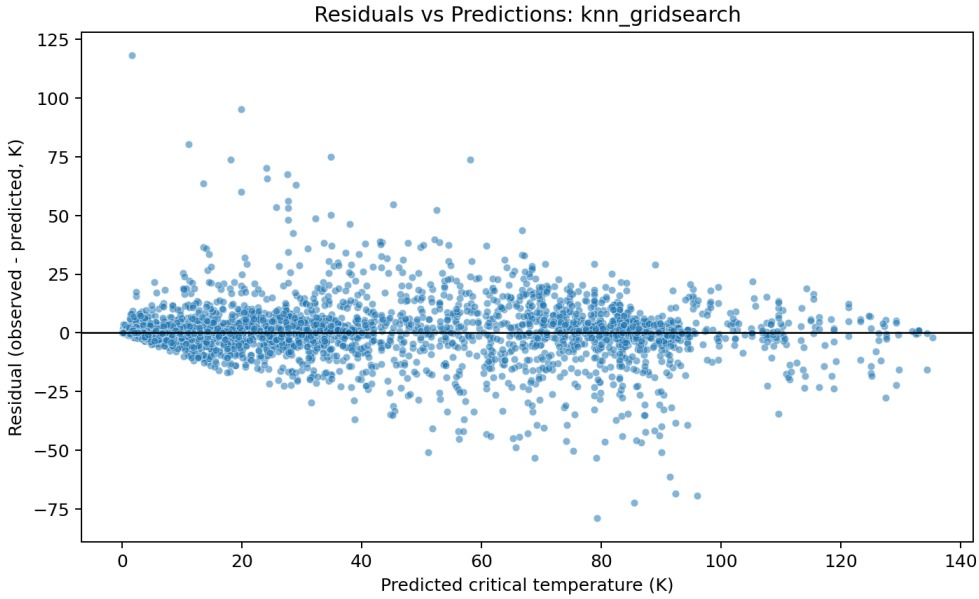

Figure 8: Residuals of the selected model as a function of predicted critical temperature. This diagnostic checks whether large errors are concentrated in specific prediction ranges.

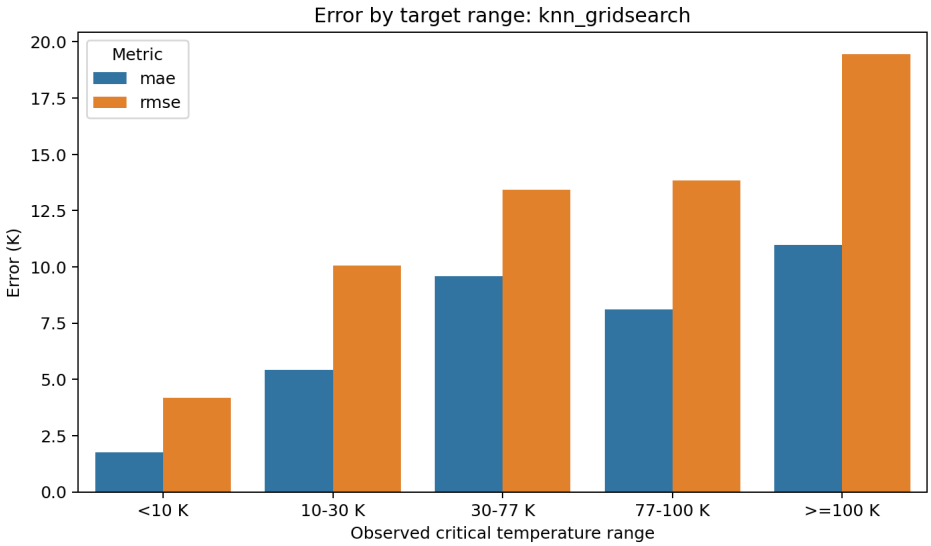

Figure 9: MAE and RMSE of the selected model by observed target range. The highest-temperature materials are the most difficult for the model.

8 Discussion

The comparison leads to two main conclusions. First, the nonlinear structure in the dataset is strong: k-NN and gradient boosting improve the RMSE by more than 7 K compared with the linear methods. Second, the best predictive model is not necessarily the most interpretable model. Ridge and Lasso summarize the marginal linear contribution of standardized descriptors and are easier to explain, but both lose substantial predictive accuracy. k-NN gives the best held-out metrics in this run, while gradient boosting is nearly as accurate and may be easier to use in production because prediction does not require searching through all training observations.

The target transformation also matters. Fitting on $\log(1 + Y)$ reduces the influence of extremely high target values and helps all methods produce positive, stable predictions after inverse transformation. The trade-off is that the fitted objective is not exactly the same as minimizing squared error directly in Kelvin. This trade-off is acceptable here because final evaluation is always performed on the original scale and because the transformed target is much less skewed.

9 Reproducibility

All main computations are organized as reusable Python modules under `src/`. The scripts `scripts/run_eda.py`, `scripts/train_models.py`, and `scripts/evaluate_models.py` produce the tables, fitted model artifacts, and figures used in this report. The configuration file `configs/default.yaml` records the random seed, the train-test split, the cross-validation settings, the metric list, and model hyperparameter grids. Package versions are managed with `uv` and locked in `uv.lock`. This organization makes the final notebook a clean assembly of already-tested pipeline steps rather than the only place where the analysis exists.

10 Limitations and Possible Extensions

The final model is evaluated on a random held-out split, not on an external experimental dataset. The reported error therefore measures generalization to similar observations from the same data source, not necessarily to future materials synthesized under different experimental conditions. In addition, the analysis uses the engineered UCI descriptors directly and does not reconstruct chemical features from raw formulae. Some materials with very high critical temperature are still underpredicted, suggesting that more specialized features, local models for the high-temperature tail, or uncertainty estimates could be useful extensions.

Another limitation is interpretability. The strongest model, k-NN, is local and non-parametric. It explains a prediction through nearby training examples rather than through global coefficients. For a scientific report, Ridge and Lasso remain valuable because they provide a simpler linear reference, but their predictive performance is clearly lower. Future work could add model-agnostic interpretation tools or compare with additional methods after explaining them carefully, provided that the course-method distinction is maintained.

11 Conclusion

The project defines a supervised regression problem for predicting `critical_temp` from 81 physical and chemical covariates. After duplicate removal, leakage-controlled preprocessing, cross-validation on the training set, and final evaluation on a held-out test set, k-nearest-neighbor regression with $k = 3$ is the recommended model. It achieves RMSE about 10.66 K, MAE about 5.67 K, and R^2 about 0.904. Histogram-based gradient boosting is a very close nonlinear alternative. Linear regression, Ridge, and Lasso are useful baselines and course-connected references, but they are not accurate enough to match the nonlinear methods on this dataset. The

final diagnostic analysis shows that the main remaining challenge is accurate prediction for the high-temperature tail.